

1.6-More on Linear System and Invertible Matrix

1. Determine the values of λ for which matrix is not invertible.

a-

$$\begin{bmatrix} 1 & \lambda & 0 \\ 3 & 2 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\det A = 1 \begin{vmatrix} 2 & 0 \\ 2 & 1 \end{vmatrix} - \lambda \begin{vmatrix} 3 & 0 \\ 1 & 1 \end{vmatrix} + 0 \begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix}$$

$$\det A = 1(2-0) - \lambda(3-0) + 0$$

$$\det A = 2 - 3\lambda$$

$$0 = 2 - 3\lambda \Rightarrow 3\lambda = 2 \Rightarrow \lambda = \frac{2}{3}$$

b-

$$\begin{bmatrix} 1 & \lambda & 0 \\ 1 & 3 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

↳ same as a

c-

$$\begin{bmatrix} 2 & \lambda & 1 \\ 3 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

same as a.

2. Determine values of λ for which A is invertible
the ~~find~~ those value in part a find inverse of A

$$\begin{bmatrix} 1 & \lambda & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det A = 1 \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} + \kappa \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} + 0 \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix}$$

$$\det A = 1(1-0) + \kappa(1-0) + 0$$

$$\det A = 1 + \kappa$$

$$0 \neq 1 - \kappa \Rightarrow$$

$$\boxed{\kappa \neq 1}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & \kappa & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & \kappa & 0 & 1 & 0 & 0 \\ 0 & 1-\kappa & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] R_2 - R_1$$

$$\begin{array}{l} R_2 \\ 1-\kappa \end{array} \left[\begin{array}{ccc|ccc} 1 & \kappa & 0 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{1-\kappa} & -\frac{1}{1-\kappa} & \frac{1}{1-\kappa} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & \kappa & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -\frac{1}{1-\kappa} & \frac{1}{1-\kappa} & -\frac{1}{1-\kappa} \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 - \\ (\frac{1}{1-\kappa})R_3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{1-\kappa} & -\frac{\kappa}{1-\kappa} & \frac{\kappa}{1-\kappa} \\ 0 & 1 & 0 & -\frac{1}{1-\kappa} & \frac{1}{1-\kappa} & -\frac{1}{1-\kappa} \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] R_1 - \kappa R_2$$

$$A^{-1} = \begin{bmatrix} \frac{1}{1-\kappa} & -\frac{\kappa}{1-\kappa} & \frac{\kappa}{1-\kappa} \\ -\frac{1}{1-\kappa} & \frac{1}{1-\kappa} & -\frac{1}{1-\kappa} \\ 0 & 0 & 1 \end{bmatrix} \quad \text{Ans.}$$

Determine those values of λ for which matrix is invertible.

$$\begin{bmatrix} \lambda & -1 & 0 \\ -1 & \lambda & -1 \\ 0 & -1 & \lambda \end{bmatrix}$$

same..

Example 1:- (Solution of Linear system using A^{-1})

$$x_1 + 2x_2 + 3x_3 = 5$$

$$2x_1 + 5x_2 + 3x_3 = 3$$

$$x_1 + 8x_3 = 17$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad B = \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix}$$

$$A X = B$$

$$X = A^{-1} B$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \begin{array}{l} R_1 - 2R_2 \\ R_3 + 2R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 9 & -3 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \xrightarrow{-R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -10 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \begin{array}{l} R_1 + 3R_3 \\ R_2 + 3R_3 \end{array}$$

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

$$X = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix}$$

$$X = \begin{bmatrix} -200 + 48 + 153 \\ 65 - 15 - 51 \\ +25 - 6 - 17 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

Ans.

$$x_1 = 1$$

$$x_2 = -1$$

$$x_3 = 2$$

Example: 2 (Using Row operation Find A^{-1})

converted in
Example 1

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

Example: 4 (Solve Linear System at once)

$$(a) \begin{aligned} x_1 + 2x_2 + 3x_3 &= 4 \\ 2x_1 + 5x_2 + 3x_3 &= 5 \\ x_1 + 8x_3 &= 9 \end{aligned}$$

$$(b) \begin{aligned} x_1 + 2x_2 + 3x_3 &= 1 \\ 2x_1 + 5x_2 + 3x_3 &= 6 \\ x_1 + 8x_3 &= -6 \end{aligned}$$

$$\left[\begin{array}{ccc|c|c} 1 & 2 & 3 & 4 & 1 \\ 2 & 5 & 3 & 5 & 6 \\ 1 & 0 & 8 & 9 & -6 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c|c} 1 & 2 & 3 & 4 & 1 \\ 0 & 1 & -3 & -3 & 4 \\ 0 & -2 & 5 & 5 & -7 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c|c} 1 & 0 & 9 & 10 & -7 \\ 0 & 1 & -3 & -3 & 4 \\ 0 & 0 & -1 & 1 & -15 \end{array} \right] \begin{array}{l} R_1 - 2R_2 \\ R_3 + 2R_2 \end{array}$$

$$\left[\begin{array}{ccc|c|c} 1 & 0 & 9 & 10 & -7 \\ 0 & 1 & -3 & -3 & 4 \\ 0 & 0 & 1 & 1 & -1 \end{array} \right] \begin{array}{l} -9R_3 \end{array}$$

$$\left[\begin{array}{ccc|c|c} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 \end{array} \right] \begin{array}{l} R_2 + 3R_3 \\ R_1 - 9R_3 \end{array}$$

$$x_1 = 1, x_2 = 0, x_3 = 1$$

$$x_1 = 2, x_2 = 1, x_3 = -1$$

Example: 3 (Determining Consistency by Elem. Method)

$$x_1 + x_2 + 2x_3 = b_1$$

$$x_1 + x_3 = b_2$$

$$2x_1 + x_2 + 3x_3 = b_3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 1 & 0 & 1 & b_2 \\ 2 & 1 & 3 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & -1 & -1 & b_2 - b_1 \\ 0 & -1 & -1 & b_3 - 2b_1 \end{array} \right] \begin{array}{l} R_2 - R_1 \\ R_3 - 2R_1 \end{array}$$

$$-R_2 \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 1 & b_1 - b_2 \\ 0 & -1 & -1 & b_3 - 2b_1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & b_2 \\ 0 & 1 & 1 & b_1 - b_2 \\ 0 & 0 & 0 & b_3 - b_1 - b_2 \end{array} \right] \begin{array}{l} R_1 - R_2 \\ R_3 + R_2 \end{array}$$

For consistency

$$b_3 - b_2 - b_1 = 0$$

$$b_3 = b_1 + b_2$$

Example : 4 (determine consistency by Elim.)

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= b_1 \\ 2x_1 + 5x_2 + 3x_3 &= b_2 \\ x_1 + 8x_3 &= b_3 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 2 & 5 & 3 & b_2 \\ 1 & 0 & 8 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 0 & 1 & -3 & b_2 - 2b_1 \\ 0 & 2 & 5 & b_3 - b_1 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 9 & 5b_1 - 2b_2 \\ 0 & 1 & -3 & b_2 - 2b_1 \\ 0 & 0 & -1 & b_3 + 2b_2 - 5b_1 \end{array} \right] \begin{array}{l} R_1 - 2R_2 \rightarrow \\ R_3 + 2R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 9 & 5b_1 - 2b_2 \\ 0 & 1 & -3 & b_2 - 2b_1 \\ 0 & 0 & 1 & 5b_1 - 2b_2 - b_3 \end{array} \right]$$

Consistent For all value. because there is no zero row.

Exercise set 1.6

Question: 1-8

Solve system by inverting coefficient matrix and using Theorem 1.6.2.

6-

$$-x - 2y - 3z = 0$$

$$w + x + 4y + 4z = 7$$

$$w + 3x + 7y + 9z = 4$$

$$-w - 2x - 4y - 6z = 6$$

$$\left[\begin{array}{cccc|cccc} 0 & -1 & -2 & -3 & 1 & 0 & 0 & 0 \\ 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 1 & 3 & 7 & 9 & 0 & 0 & 1 & 0 \\ -1 & -2 & -4 & -6 & 0 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 0 & -1 & -2 & -3 & 1 & 0 & 0 & 0 \\ 1 & 3 & 7 & 9 & 0 & 0 & 1 & 0 \\ -1 & -2 & -4 & -6 & 0 & 0 & 0 & 1 \end{array} \right] \quad R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 0 & -1 & -2 & -3 & 1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 5 & 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & -2 & 0 & 1 & 0 & 1 \end{array} \right] \quad \begin{array}{l} R_3 - R_1 \\ R_4 + R_1 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 5 & 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & -2 & 0 & 1 & 0 & 1 \end{array} \right] \quad -R_2$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 & 0 & 1 \end{array} \right] \begin{array}{l} R_1 - R_2 \\ R_3 - 2R_2 \\ R_4 + R_2 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 & -1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 & 0 & 1 \end{array} \right] -R_3$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & -1 & 5 & -1 & 2 & 0 \\ 0 & 1 & 0 & 1 & 3 & -2 & 2 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 & 3 & -1 & 2 & 1 \end{array} \right] \begin{array}{l} R_1 - 2R_3 \\ R_2 - 2R_3 \\ R_4 - 2R_3 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & -1 & 5 & -1 & 2 & 0 \\ 0 & 1 & 0 & 1 & 3 & -2 & 2 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -3 & 1 & -2 & -1 \end{array} \right] -R_4$$

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 2 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 6 & -3 & 4 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -3 & 1 & -2 & -1 \end{array} \right] \begin{array}{l} R_3 - R_4 \\ R_2 - R_4 \\ R_1 + R_4 \end{array}$$

$$A^{-1} = \begin{bmatrix} 2 & 0 & 0 & -1 \\ 6 & -3 & 4 & 1 \\ 1 & 0 & 1 & 1 \\ -3 & 1 & -2 & -1 \end{bmatrix}$$

$$X = A^{-1}b$$

$$X = \begin{bmatrix} 2 & 0 & 0 & -1 \\ 6 & -3 & 4 & 1 \\ 1 & 0 & 1 & 1 \\ -3 & 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 7 \\ 4 \\ 6 \end{bmatrix}$$

$$X = \begin{bmatrix} 0+0+0-6 \\ 0-21+16+6 \\ 0+0+4+6 \\ 0+7-8-6 \end{bmatrix} \Rightarrow \begin{bmatrix} -6 \\ 1 \\ 10 \\ -7 \end{bmatrix}$$

$$w = -6, \quad x = 1, \quad y = 10, \quad z = -7$$

8-

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= b_1 \\ 2x_1 + 5x_2 + 5x_3 &= b_2 \\ 3x_1 + 5x_2 + 8x_3 &= b_3 \end{aligned}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 5 & 0 & 1 & 0 \\ 3 & 5 & 8 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & -1 & -1 & -3 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - 3R_1 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 5 & 5 & -2 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & -2 & -5 & +1 & 1 \end{array} \right] \begin{array}{l} R_1 - 2R_2 \\ R_3 + R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 5 & 5 & -2 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5/2 & -1/2 & -1/2 \end{array} \right] \begin{array}{l} \\ \\ -\frac{1}{2}R_3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{15}{2} & \frac{1}{2} & \frac{5}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & -1 & \frac{5}{2} & -\frac{1}{2} & -\frac{1}{2} \end{array} \right] \begin{array}{l} R_2 + R_3 \\ R_1 - 5R_3 \end{array}$$

$$A^{-1} = \begin{bmatrix} -\frac{15}{2} & \frac{1}{2} & \frac{5}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{5}{2} & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$X = A^{-1}b$$

$$X = \begin{bmatrix} -\frac{15}{2} & \frac{1}{2} & \frac{5}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{5}{2} & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$X = \begin{bmatrix} -\frac{15}{2}b_1 + \frac{1}{2}b_2 + \frac{5}{2}b_3 \\ \frac{1}{2}b_1 + \frac{1}{2}b_2 - \frac{1}{2}b_3 \\ \frac{5}{2}b_1 - \frac{1}{2}b_2 - \frac{1}{2}b_3 \end{bmatrix}$$

$$X = \begin{bmatrix} \frac{-15b_1 + b_2 + 5b_3}{2} \\ \frac{b_1 + b_2 - b_3}{2} \\ \frac{5b_1 - b_2 - b_3}{2} \end{bmatrix}$$

Ans.

1, 2, 3, 4, 5, 7 : are same.

Question : 9-12

11-

$$4x_1 - 7x_2 = b_1$$

$$x_1 + 2x_2 = b_2$$

(i) $b_1 = 1, b_2 = 1$

(ii) $b_1 = -4, b_2 = 6$

(iii) $b_1 = -1, b_2 = 3$

(iv) $b_1 = -5, b_2 = 1$

$$\left[\begin{array}{cc|c|c|c|c} 4 & -7 & 1 & -4 & -1 & -5 \\ 1 & 2 & 1 & 6 & 3 & 1 \end{array} \right]$$

$$\left[\begin{array}{cc|c|c|c|c} 1 & 2 & 1 & 6 & 3 & 1 \\ 4 & -7 & 1 & -4 & -1 & -5 \end{array} \right] R_2 \leftrightarrow R_1$$

$$\left[\begin{array}{cc|c|c|c|c} 1 & 2 & 1 & 6 & 3 & 1 \\ 0 & -15 & -4 & -28 & -13 & -9 \end{array} \right] R_2 - 4R_1$$

$$\left[\begin{array}{cc|c|c|c|c} 1 & 2 & 1 & 6 & 3 & 1 \\ 0 & 1 & +\frac{4}{15} & \frac{28}{5} & \frac{13}{15} & \frac{3}{5} \end{array} \right]$$

$$\frac{1}{15} R_2$$

$$\left[\begin{array}{cc|c|c|c|c} 1 & 0 & \frac{7}{15} & \frac{34}{15} & \frac{19}{15} & -\frac{1}{5} \\ 0 & 1 & \frac{4}{15} & \frac{28}{5} & \frac{13}{15} & \frac{3}{5} \end{array} \right] R_1 - 2R_2$$

(i) $x_1 = \frac{7}{15}, x_2 = \frac{4}{5}$

(ii) $x_1 = \frac{34}{15}, x_2 = \frac{28}{15}$

(iii) $x_1 = \frac{19}{15}, x_2 = \frac{13}{15}$

(iv) $x_1 = -\frac{1}{5}, x_2 = \frac{3}{5}$

12-

$$x_1 + 3x_2 + 5x_3 = b_1$$

$$-x_1 - 2x_2 = b_2$$

$$2x_1 + 5x_2 + 4x_3 = b_3$$

(i) $b_1 = 1, b_2 = 0, b_3 = -1$

(ii) $b_1 = 0, b_2 = 1, b_3 = 1$

(iii) $b_1 = -1, b_2 = -1, b_3 = 0$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 5 & 1 & 0 & -1 \\ -1 & -2 & 0 & 0 & 1 & -1 \\ 2 & 5 & 4 & -1 & 1 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 5 & 1 & 0 & -1 \\ 0 & 1 & 5 & 1 & 1 & -2 \\ 0 & -1 & -6 & -3 & 1 & 2 \end{array} \right] \begin{array}{l} R_2 + R_1 \\ R_3 - 2R_1 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -10 & -2 & -3 & 5 \\ 0 & 1 & 5 & 1 & 1 & -2 \\ 0 & 0 & -1 & -2 & 2 & 0 \end{array} \right] \begin{array}{l} R_1 - 3R_2 \\ R_3 + R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -10 & -2 & -3 & 5 \\ 0 & 1 & 5 & 1 & 1 & -2 \\ 0 & 0 & 1 & 2 & -2 & 0 \end{array} \right] -R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 18 & -23 & 5 \\ 0 & 1 & 0 & -9 & 11 & -2 \\ 0 & 0 & 1 & 2 & -2 & 0 \end{array} \right] \begin{array}{l} R_2 - 5R_3 \\ R_1 + 10R_3 \end{array}$$

(i) $x_1 = 18, x_2 = -9, x_3 = 2$ (ii) $x_1 = -23, x_2 = 11, x_3 = -2$

(iii) $x_1 = 5, x_2 = 2, x_3 = 0$

9, 10, are also same.

Question: 13-17

Determine conditions on b_i if any, in order to guarantee the linear system is consistent.

15-

$$\begin{aligned}x_1 - 2x_2 + 5x_3 &= b_1 \\4x_1 - 5x_2 + 8x_3 &= b_2 \\-3x_1 + 3x_2 - 3x_3 &= b_3\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 5 & b_1 \\ 4 & -5 & 8 & b_2 \\ -3 & 3 & -3 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 5 & b_1 \\ 0 & 3 & -12 & b_2 - 4b_1 \\ 0 & -3 & 12 & b_3 + 3b_1 \end{array} \right] \begin{array}{l} R_2 - 4R_1 \\ R_3 + 3R_1 \end{array}$$

$$\begin{array}{l} R_2 \\ 3 \end{array} \left[\begin{array}{ccc|c} 1 & -2 & 5 & b_1 \\ 0 & 1 & -4 & \frac{b_2 - 4b_1}{3} \\ 0 & -3 & 12 & b_3 + 3b_1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 5 & b_1 \\ 0 & 1 & -4 & \frac{b_2 - 4b_1}{3} \\ 0 & 0 & 0 & \frac{2(b_3 - b_1 + b_2)}{3} \end{array} \right] \begin{array}{l} \\ \\ R_3 + 3R_2 \end{array}$$

$$\begin{aligned}b_3 - b_1 + b_2 &= 0 \\ b_3 &= b_1 - b_2\end{aligned}$$

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$$\begin{aligned}x_1 - 2x_2 - x_3 &= b_1 \\-4x_1 + 5x_2 + 2x_3 &= b_2 \\-4x_1 + 7x_2 + 4x_3 &= b_3\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ -4 & 5 & 2 & b_2 \\ -4 & 7 & 4 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & -3 & -2 & b_2 + 4b_1 \\ 0 & -1 & 0 & b_3 + 4b_1 \end{array} \right] \begin{array}{l} \\ R_2 + 4R_1 \\ R_3 + 4R_1 \end{array}$$

$$R_2 \leftrightarrow R_3 \quad \left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & -1 & 0 & b_3 + 4b_1 \\ 0 & -3 & -2 & b_2 + 4b_1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & 1 & 0 & -b_3 - 4b_1 \\ 0 & -3 & -2 & b_2 + 4b_1 \end{array} \right] -R_2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & 1 & 0 & -b_3 - 4b_1 \\ 0 & 0 & -2 & b_2 - 8b_1 - 3b_3 \end{array} \right] R_3 + 3R_2$$

Every row has pivot

so $b_i \in \mathbb{R}$

17-

$$\begin{aligned} x_1 - x_2 + 3x_3 + 2x_4 &= b_1 \\ -2x_1 + x_2 + 5x_3 + x_4 &= b_2 \\ -3x_1 + 2x_2 + 2x_3 - x_4 &= b_3 \\ 4x_1 + 3x_2 + x_3 + 3x_4 &= b_4 \end{aligned}$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 3 & 2 & b_1 \\ -2 & 1 & 5 & 1 & b_2 \\ -3 & 2 & 2 & -1 & b_3 \\ 4 & -3 & 1 & 3 & b_4 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 3 & 2 & b_1 \\ 0 & -1 & 11 & 5 & b_2 + 2b_1 \\ 0 & -1 & 11 & 5 & b_3 + 3b_1 \\ 0 & 1 & -11 & -5 & b_4 - 4b_1 \end{array} \right] \begin{array}{l} R_2 + 2R_1 \\ R_3 + 3R_1 \\ R_4 - 4R_1 \end{array}$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 3 & 2 & b_1 \\ 0 & 1 & -11 & -5 & -b_2 - 2b_1 \\ 0 & -1 & 11 & 5 & b_3 + 3b_1 \\ 0 & 1 & -11 & -5 & b_4 - 4b_1 \end{array} \right] -R_2$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 3 & 2 & b_1 \\ 0 & 1 & -11 & -5 & -b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 + b_1 \\ 0 & 0 & 0 & 0 & b_4 - b_3 - 7b_1 \end{array} \right] \begin{array}{l} R_3 + R_2 \\ R_4 - R_2 \end{array}$$

$$b_3 - b_2 + b_1 = 0$$

$$b_3 = b_2 - b_1$$

$$b_4 + b_2 - 2b_1 = 0$$

$$b_4 = 2b_1 - b_2$$

13 & 14

are same

Question : 19-20

19-

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & 3 & 0 \\ 0 & 2 & -1 \end{bmatrix} x = \begin{bmatrix} 2 & -1 & 5 & 7 & 8 \\ 4 & 0 & -3 & 0 & 1 \\ 3 & 5 & -7 & 2 & 1 \end{bmatrix} \begin{array}{c} \\ \\ b \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 2 & -1 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 5 & -2 & -2 & 1 & 0 \\ 0 & 2 & -1 & 0 & 0 & 1 \end{array} \right] R_2 - 2R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & -2 \\ 0 & 2 & -1 & 0 & 0 & 1 \end{array} \right] R_2 - 2R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -1 & 1 & -2 \\ 0 & 1 & 0 & -2 & 1 & -2 \\ 0 & 0 & -1 & 4 & -2 & 5 \end{array} \right] \begin{array}{l} R_1 + R_2 \\ R_3 - 2R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -1 & 1 & -2 \\ 0 & 1 & 0 & -2 & 1 & -2 \\ 0 & 0 & 1 & -4 & 2 & -5 \end{array} \right] \xrightarrow{-R_3}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & 3 \\ 0 & 1 & 0 & -2 & 1 & -2 \\ 0 & 0 & 1 & -4 & 2 & -5 \end{array} \right] \xrightarrow{R_1 - R_3}$$

$$X = A^{-1}b$$

$$X = \begin{bmatrix} 3 & -1 & 3 \\ -2 & 1 & -2 \\ -4 & 2 & -5 \end{bmatrix} \begin{bmatrix} 2 & -1 & 5 & 7 & 8 \\ 4 & 0 & -3 & 0 & 1 \\ 3 & 5 & -7 & 2 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 6-4+9 & -3+0+15 & 15+3-21 & 21-0+6 & 24-1+3 \\ -4+4-6 & 2+0-10 & -10-3+4 & -4+0-4 & -16+1-2 \\ -8+8-15 & 4+0-25 & -20-6+35 & -18+0-10 & -32+2-5 \end{bmatrix}$$

$$X = \begin{bmatrix} 11 & 12 & -3 & 27 & 26 \\ -6 & -8 & 1 & -18 & -17 \\ -15 & -21 & 9 & -38 & -35 \end{bmatrix} \quad \text{Ans}$$

20-

$$\begin{bmatrix} -2 & 0 & 1 \\ 0 & -1 & -1 \\ 1 & 1 & -4 \end{bmatrix} X = \begin{bmatrix} 4 & 3 & 2 & 1 \\ 6 & 7 & 8 & 9 \\ 1 & 3 & 7 & 9 \end{bmatrix}$$

A X B

$$\left[\begin{array}{ccc|ccc} -2 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 1 & 1 & -4 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -4 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ -2 & 0 & 1 & 1 & 0 & 0 \end{array} \right] \quad R_1 \leftrightarrow R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -4 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 0 & 2 & -7 & 1 & 0 & 2 \end{array} \right] \quad R_3 + 2R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -4 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 2 & -7 & 1 & 0 & 2 \end{array} \right] \quad -R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -5 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & -9 & 1 & 2 & 2 \end{array} \right] \quad \begin{array}{l} R_1 - R_2 \\ R_3 - 2R_1 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -5 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -\frac{1}{9} & -\frac{2}{9} & -\frac{2}{9} \end{array} \right] \quad -\frac{1}{9} R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{5}{9} & -\frac{1}{9} & -\frac{1}{9} \\ 0 & 1 & 0 & \frac{1}{9} & -\frac{7}{9} & \frac{2}{9} \\ 0 & 0 & 1 & -\frac{1}{9} & -\frac{2}{9} & -\frac{2}{9} \end{array} \right] \quad \begin{array}{l} R_2 - R_3 \\ R_1 + 5R_3 \end{array}$$

$$A^{-1} = \frac{1}{9} \begin{bmatrix} -5 & -1 & -1 \\ 1 & -7 & 2 \\ -1 & -2 & -2 \end{bmatrix}$$

$$x = A^{-1}b$$

$$x = \frac{1}{9} \begin{bmatrix} -5 & -1 & -1 \\ 1 & -7 & 2 \\ -1 & -2 & -2 \end{bmatrix} \begin{bmatrix} 4 & 3 & 2 & 1 \\ 6 & 7 & 8 & 9 \\ 1 & 3 & 7 & 9 \end{bmatrix}$$

$$X = \frac{1}{9} \begin{bmatrix} -20-6-1 & -15-7-3 & -10-8-7 & -5-9-9 \\ 4-42+2 & 3-49+6 & 2-56+14 & 1-63+18 \\ -4-12-2 & -3-14-6 & -2-16-14 & -1-18-18 \end{bmatrix}$$

$$X = \frac{1}{9} \begin{bmatrix} -27 & -25 & -25 & -23 \\ -36 & -40 & -40 & -44 \\ -18 & -23 & -32 & -37 \end{bmatrix} \text{ Ang}$$